

FourierGNN: Rethinking Multivariate Time Series Forecasting from a Pure Graph Perspective

Qualitative and Quantitative Analysis of Results / Contributions

1 Quantitative Analysis

- **Overall Performance:** Our reproduced runs show mixed results; some datasets (e.g. Solar, METR-LA) achieve comparable or improved MAE/RMSE, while others (Electricity, Traffic, Wiki) show substantial degradation in one or more metrics.
- **MAPE vs MAE/RMSE mismatch:** We observe cases where MAE/RMSE are better but MAPE is much worse (notably COVID-19, Wiki). This occurs because MAPE is a relative error metric that inflates when ground-truth values are small or when inverse scaling differs across runs.
- **Training time and efficiency:** Measured per-epoch times are higher for several models, likely cause of difference in GPU and/or CUDA settings.

Dataset	MAE		RMSE		MAPE (%)		Time/Epoch (s)	
	Paper	Ours	Paper	Ours	Paper	Ours	Paper	Ours
ECG	0.052	0.057	0.078	0.089	10.97	13.52	—	2.94
COVID-19	0.123	0.080	0.168	0.129	71.52	137.36	—	0.75
SOLAR	0.120	0.106	0.162	0.149	116.48	61.88	—	22.56
TRAFFIC	0.011	0.082	0.023	0.125	28.71	79.07	99.25	127.30
ELECTRICITY	0.051	2.739	0.077	9.454	24.28	200.20	—	122.55
METR-LA	0.084	0.085	0.165	0.168	89.69	49.20	—	31.31
WIKI	0.041	0.060	0.076	1.270	64.50	123.51	16.28	7.49

Table 1: Quantitative Comparison

2 Primary Quantitative Failure Modes

- **Scale and Preprocessing Mismatch:** Small differences in normalisation or inverse-transform (per-series vs global scaling, rounding) cause large changes in MAPE and can also alter RMSE slightly.
- **Data-split / Distribution Shift:** A different chronological split or inclusion/exclusion of near-zero samples changes per-dataset statistics and thus relative error measures.

3 Scaling Analysis

Upon analysis of prediction stability over increasing lengths (3, 6, 9, 12 steps), Across datasets, the model’s absolute errors (MAE, RMSE) generally increase slowly with length, as expected, indicating reasonable temporal stability. However, relative-error behaviour (MAPE) and RMSE magnitudes vary strongly by dataset, suggesting that differences in preprocessing, target scale, or implementation (inverse-scaling) are driving most discrepancies.

Length	MAE		RMSE		MAPE(%)	
	Paper	Ours	Paper	Ours	Paper	Ours
3	0.071	0.065	0.103	0.112	61.02	133.52
6	0.093	0.065	0.131	0.109	65.72	134.05
9	0.109	0.074	0.148	0.119	69.59	137.75
12	0.123	0.080	0.168	0.129	71.52	137.36

Table 2: Accuracy comparison under different prediction lengths on COVID-19 dataset

Length	MAE		RMSE		MAPE(%)	
	Paper	Ours	Paper	Ours	Paper	Ours
3	0.040	0.054	0.075	1.333	58.18	100.57
6	0.041	0.056	0.075	1.306	60.43	104.92
9	0.041	0.057	0.076	1.274	60.95	113.52
12	0.041	0.060	0.076	1.270	64.50	123.51

Table 3: Accuracy comparison under different prediction lengths on Wiki dataset

Length	MAE		RMSE		MAPE(%)	
	Paper	Ours	Paper	Ours	Paper	Ours
3	0.050	0.053	0.113	0.116	86.30	44.78
6	0.066	0.068	0.140	0.140	87.97	46.96
9	0.076	0.075	0.159	0.155	88.99	47.80
12	0.084	0.084	0.165	0.168	89.69	49.20

Table 4: Accuracy comparison under different prediction lengths on METR-LA dataset

4 Conclusion

The study reproduces and analyses FourierGNN across multiple datasets and prediction lengths, confirming its strengths in modelling periodic and moderate-dimensional time series. While MAE and RMSE trends largely align with the original paper, discrepancies—particularly in MAPE and high-dimensional settings—are primarily attributable to preprocessing, scaling, and configuration differences rather than architectural limitations. Overall, the results validate FourierGNN’s robustness and efficiency, and indicate that careful normalisation and hyperparameter tuning are critical for faithful reproduction of reported performance.

5 Contributions

Ayush Rawat - Report, Dataset Sourcing, Code

Omkar Bhandare - Code, Future Directions

Shivam Choudhury - Dataset Sourcing, Future Directions, Code